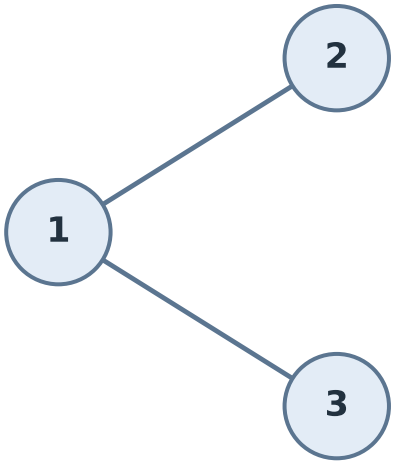
M19-01 — Build an Adjacency List

Difficulty: Easy | Topic: Graphs





1: 2 3

2: 1

3: 1

4: 5

5: 4

Sample input

```
5 3
1 3
1 2
4 5
```

Sample output

```
1: 2 3
2: 1
3: 1
4: 5
5: 4
```

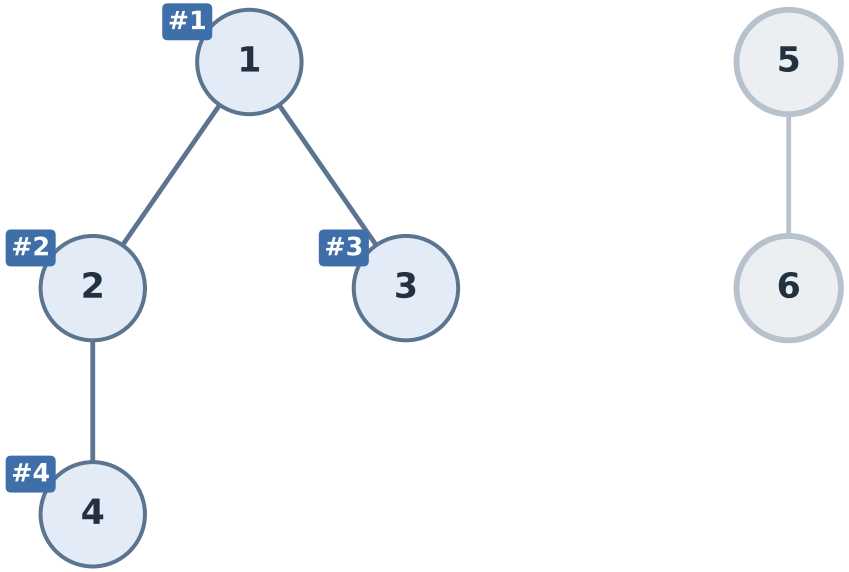
The graph and its adjacency list. Each undirected edge appears in both endpoints' lists, sorted ascending.

Walkthrough

Edge 1–3 adds 3 to 1's list and 1 to 3's list; edge 1–2 does the same for 1 and 2. After sorting, vertex 1 lists 2 3. Vertices 4 and 5 form a separate component.

M19-02 — Breadth-First Search (BFS) Traversal

Difficulty: Easy | Topic: Graphs



visit order unreachable from 1

Blue badges give the BFS visit order from S = 1. Grey vertices cannot be reached from the source.

Sample input

```
6 4
1 2
1 3
2 4
5 6
1
```

Sample output

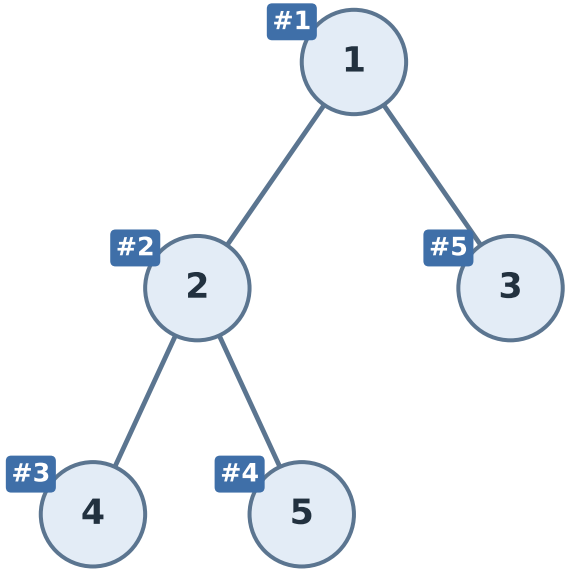
```
1 2 3 4
```

Walkthrough

BFS removes 1 from the queue and adds its neighbors 2 and 3 in ascending order; removing 2 adds 4. The queue then empties, so 5 and 6 are never visited: 1 2 3 4.

M19-03 — Depth-First Search (DFS) Traversal

Difficulty: Easy | Topic: Graphs



Sample input

```
5 4
1 2
1 3
2 4
2 5
1
```

Sample output

```
1 2 4 5 3
```

Blue badges give the order in which recursive DFS first visits each vertex, exploring neighbors in ascending order.

Walkthrough

From 1, DFS goes to its smallest neighbor 2, then to 4. Vertex 4 has nothing new, so DFS returns to 2 and visits 5, then returns to 1 and visits 3. The explicit stack keeps each vertex's position in its neighbor list to reproduce this order.

M19-04 — Check Whether an Undirected Graph Is Connected

Difficulty: Easy | Topic: Graphs



Sample input

```
4 2
1 2
3 4
```

Sample output

```
NO
```

reachable from vertex 1 not reachable

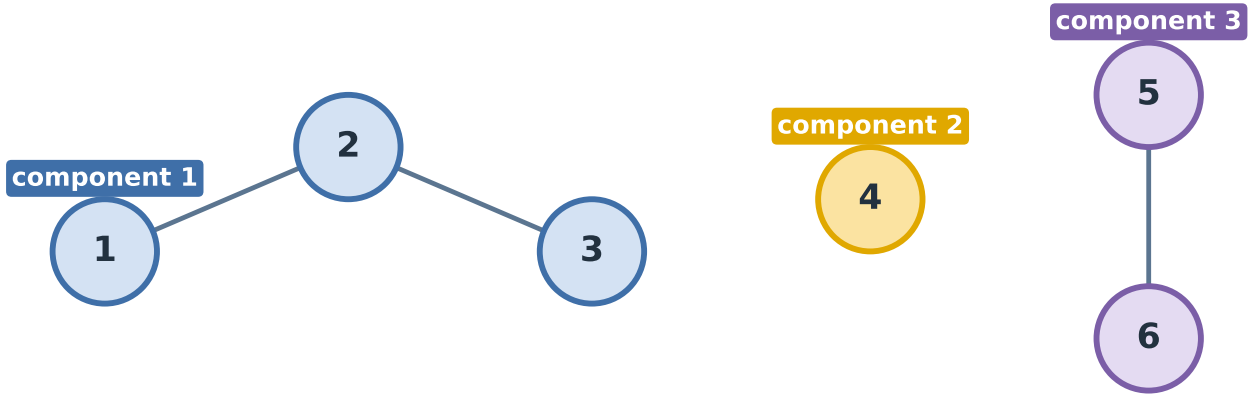
Green vertices are reached by a traversal starting at vertex 1; grey vertices are not.

Walkthrough

A BFS from 1 reaches only 2. Vertices 3 and 4 are connected to each other but not to 1, so not every vertex is reachable and the answer is NO.

M19-05 — Count Connected Components

Difficulty: Easy | Topic: Graphs



Sample input

```
6 3
1 2
2 3
5 6
```

Sample output

```
3
```

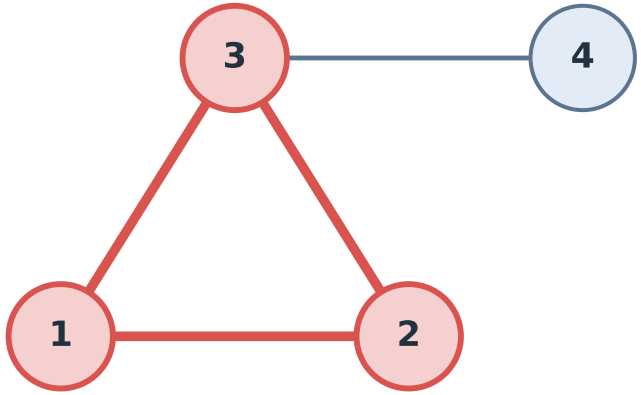
Each color is one connected component. Vertex 4 has no edges, so it forms a component by itself.

Walkthrough

The components are {1, 2, 3}, {4}, and {5, 6}. Counting a new component each time a traversal starts from an unvisited vertex (or each time a DSU union fails to merge) gives 3.

M19-06 — Detect a Cycle in an Undirected Graph

Difficulty: Medium | Topic: Graphs



edges forming the cycle

Red edges form a closed path through three distinct vertices. Edge 3–4 is a branch, not part of the cycle.

Sample input

```
4 4
1 2
2 3
3 1
3 4
```

Sample output

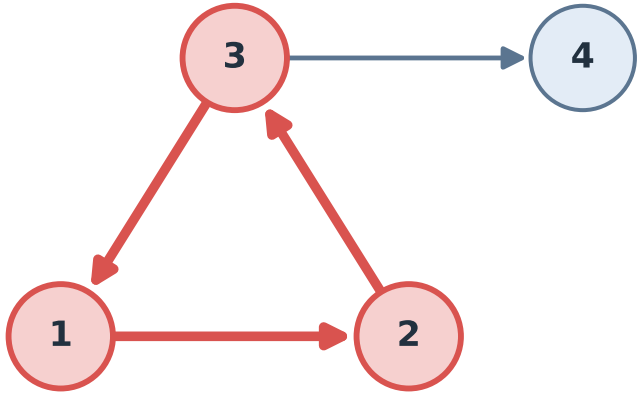
```
CYCLE
```

Walkthrough

With DSU: edges 1–2 and 2–3 join 1, 2, and 3 into one set. Edge 3–1 then connects two vertices already in the same set, so it closes the cycle 1–2–3–1.

M19-07 — Detect a Cycle in a Directed Graph

Difficulty: Medium | Topic: Graphs



directed cycle 1 → 2 → 3 → 1

Arrows show edge direction. Red arrows follow a cycle in the arrow direction.

Sample input

```
4 4
1 2
2 3
3 1
3 4
```

Sample output

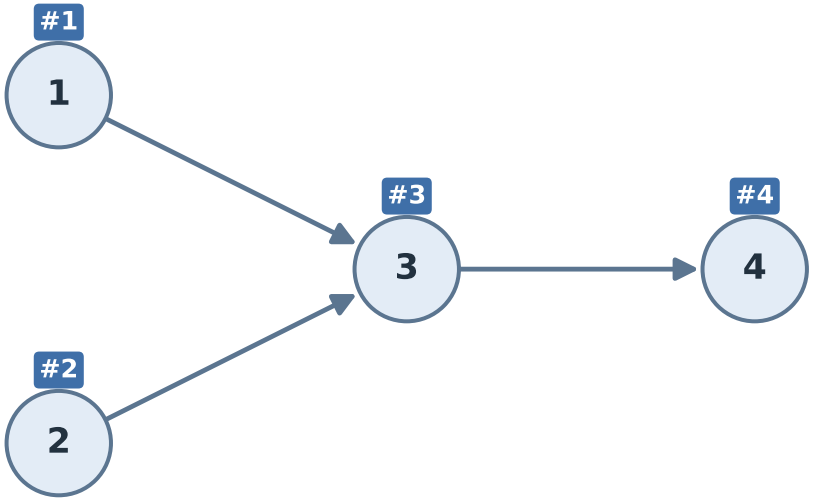
```
CYCLE
```

Walkthrough

1 → 2 → 3 → 1 returns to its start while obeying every arrow, so a directed cycle exists. In Kahn's algorithm, vertices 1, 2, and 3 never reach indegree 0.

M19-08 — Lexicographically Smallest Topological Order

Difficulty: Medium | Topic: Graphs



Sample input

```
4 3
1 3
2 3
3 4
```

Sample output

```
1 2 3 4
```

Arrows point from prerequisite to dependent. Blue badges give the lexicographically smallest topological order.

Walkthrough

Initially 1 and 2 have indegree 0; the smallest, 1, goes first, then 2. Removing both frees 3, and removing 3 frees 4: 1 2 3 4.

M19-09 — Shortest Path in an Unweighted Graph

Difficulty: Medium | Topic: Graphs



Sample input

```
5 3
1 2
2 3
4 5
1
```

Sample output

```
0 1 2 -1 -1
```

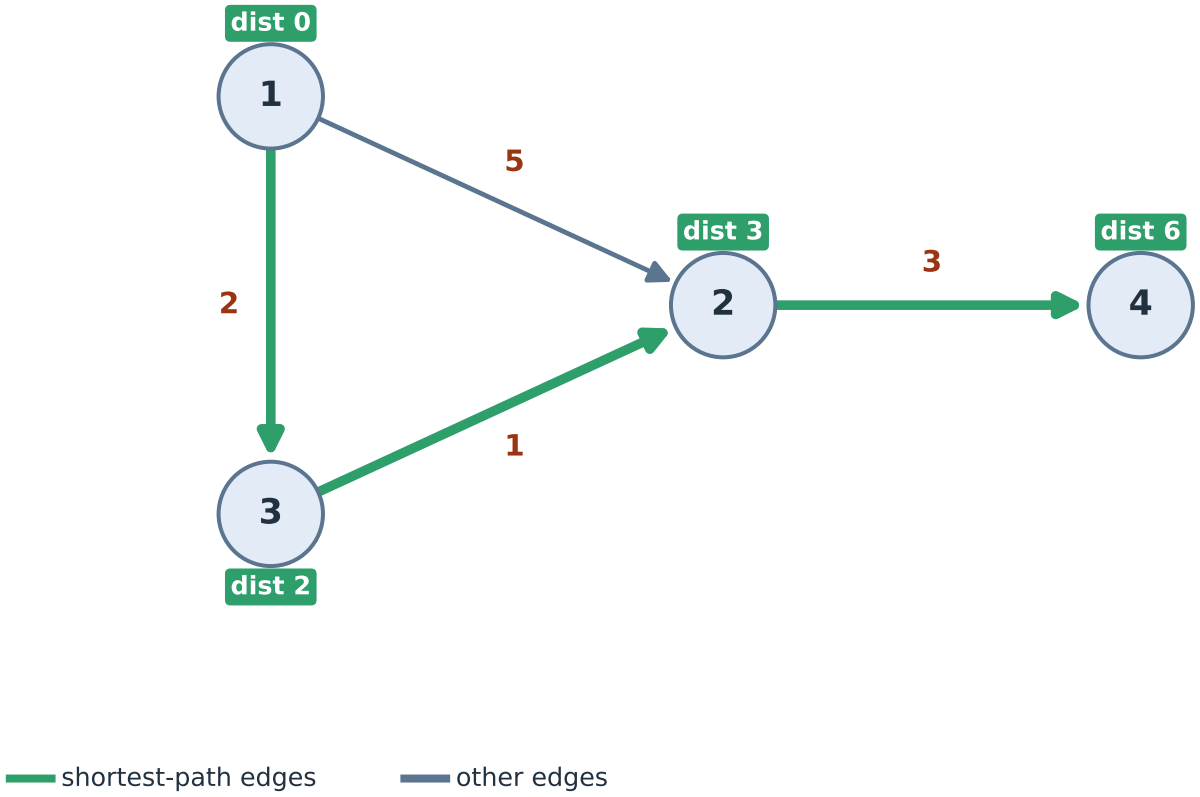
Badges show each vertex's BFS distance (in edges) from S = 1; -1 marks vertices that cannot be reached.

Walkthrough

1 is at distance 0, its neighbor 2 at 1, and 3 at 2. Vertices 4 and 5 are only connected to each other, so their distance is -1.

M19-10 — Dijkstra's Shortest Paths

Difficulty: Medium | Topic: Graphs



Sample input

```
4 4
1 2 5
1 3 2
3 2 1
2 4 3
1
```

Sample output

```
0 3 2 6
```

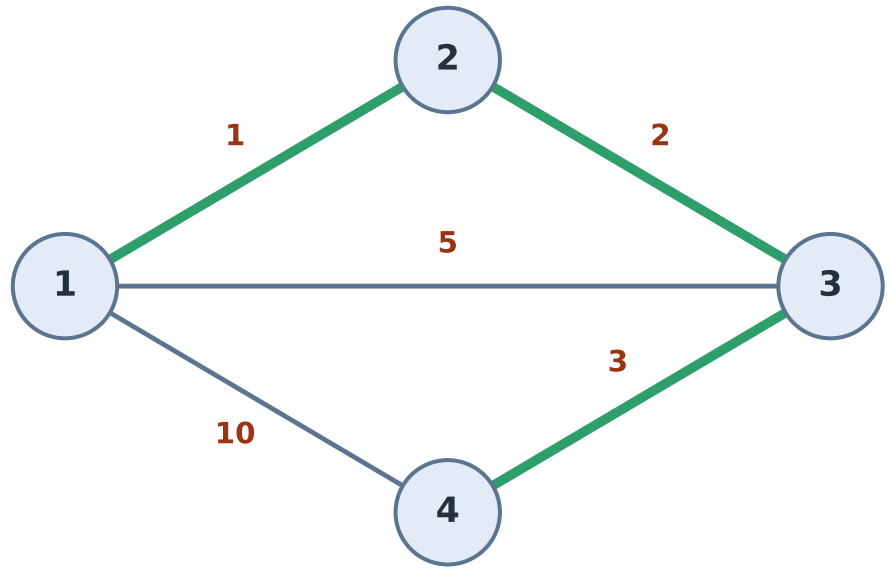
Numbers on edges are weights. Green arrows form the shortest-path tree from S = 1; badges show final distances.

Walkthrough

Dijkstra first settles 3 (distance 2). Through 3, vertex 2 costs 2 + 1 = 3, better than the direct edge of 5. Then 4 costs 3 + 3 = 6.

M19-11 — Minimum Spanning Tree Weight

Difficulty: Medium | Topic: Graphs



MST edges (1 + 2 + 3 = 6) not used

Numbers are edge weights. Green edges form the minimum spanning tree.

Sample input

```
4 5
1 2 1
2 3 2
3 4 3
1 4 10
1 3 5
```

Sample output

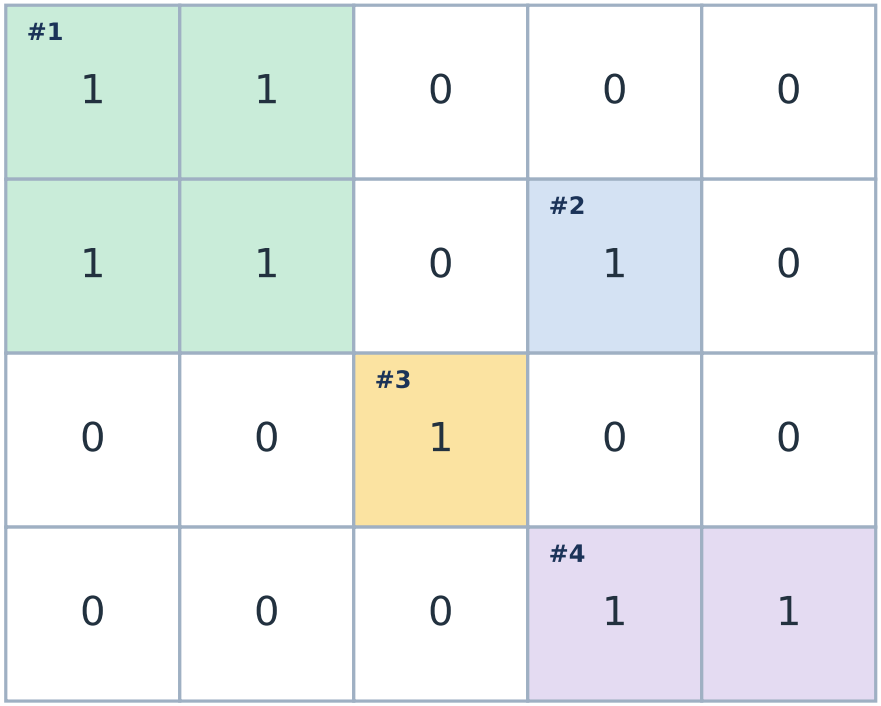
```
6
```

Walkthrough

Kruskal takes edges by weight: 1 (1–2), 2 (2–3), and 3 (3–4) each join new components. Edges 5 (1–3) and 10 (1–4) would close cycles, so they are skipped. Total 6.

M19-12 — Count Islands in a Grid

Difficulty: Medium | Topic: Graphs



Sample input

```
4 5
11000
11010
00100
00011
```

Sample output

```
4
```

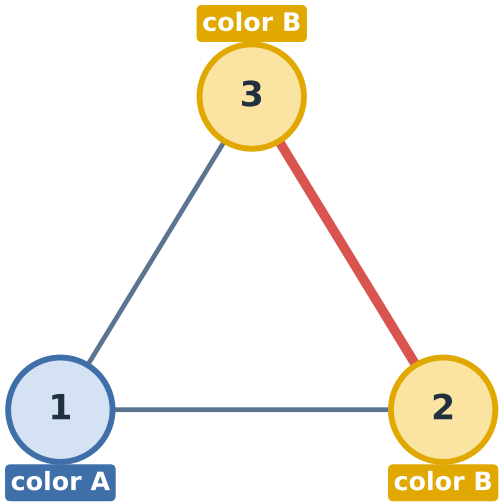
Each color is one island. Cells that touch only at a corner belong to different islands.

Walkthrough

Island 1 is the 2 × 2 block, island 2 the single cell in row 2, island 3 the single cell in row 3, and island 4 the pair in row 4. The center cells touch only diagonally, so the answer is 4.

M19-13 — Check Whether an Undirected Graph Is Bipartite

Difficulty: Medium | Topic: Graphs



Sample input

```
3 3
1 2
2 3
3 1
```

Sample output

```
NO
```

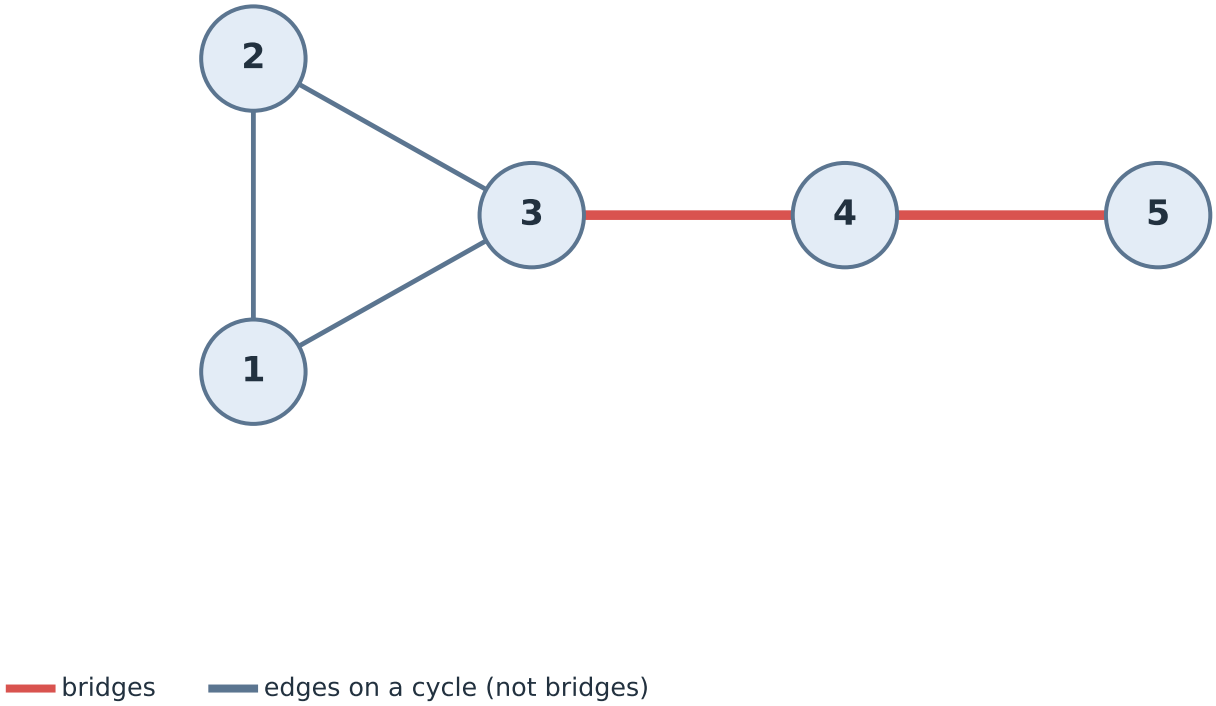
BFS colors 1 with color A and both of its neighbors, 2 and 3, with color B. The red edge 2–3 then joins two vertices of the same color.

Walkthrough

2 and 3 are both neighbors of 1, so both need color B, but 2–3 is also an edge. Whatever colors are chosen, one edge of the odd cycle ends up with matching colors, so the answer is NO.

M19-14 — Find All Bridges in an Undirected Graph

Difficulty: Hard | Topic: Graphs



Sample input

```
5 5
1 2
2 3
3 1
3 4
4 5
```

Sample output

```
2
3 4
4 5
```

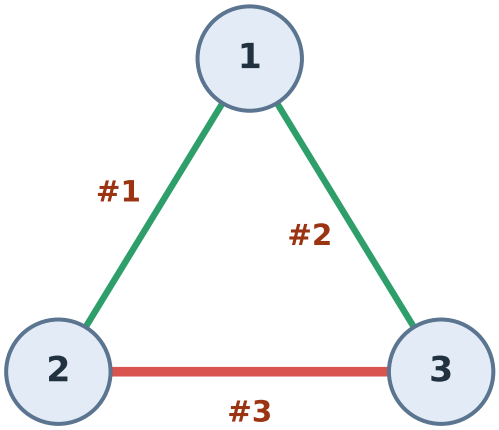
Red edges are bridges: removing either one disconnects the graph. The triangle edges each lie on a cycle.

Walkthrough

Removing 3–4 separates {4, 5} from the triangle; removing 4–5 isolates 5. Any triangle edge can be removed and its endpoints stay connected through the third vertex.

M19-15 — Find a Redundant Connection Using DSU

Difficulty: Hard | Topic: Graphs



joins two different sets

both ends already connected

Edge labels give the input order. Green edges merge two DSU sets; the red edge connects two vertices already in the same set.

Sample input

```
3
1 2
1 3
2 3
```

Sample output

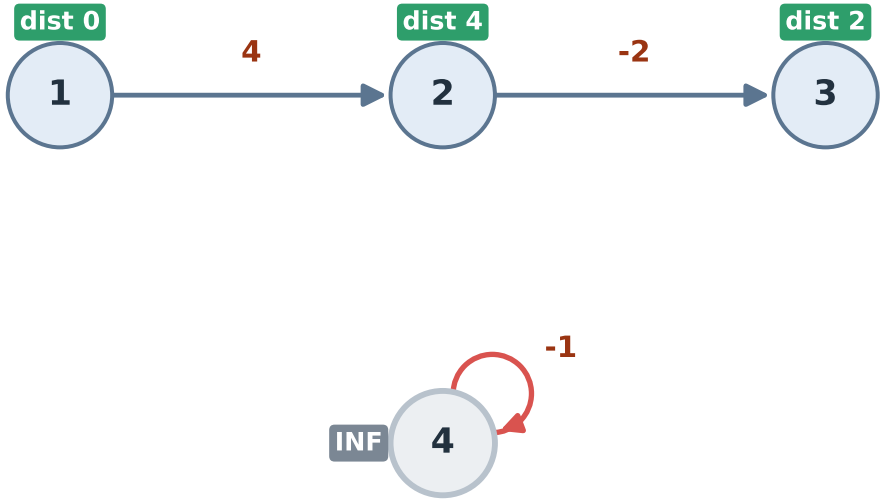
```
2 3
```

Walkthrough

Edge #1 joins 1 and 2; edge #2 joins 1 and 3, so {1, 2, 3} is one set. Edge #3 (2–3) finds both ends in that set, so it is the redundant connection: 2 3.

M19-16 — Bellman–Ford with Negative Edges

Difficulty: Hard | Topic: Graphs



Sample input

```
4 3
1 2 4
2 3 -2
4 4 -1
1
```

Sample output

```
0 4 2 INF
```

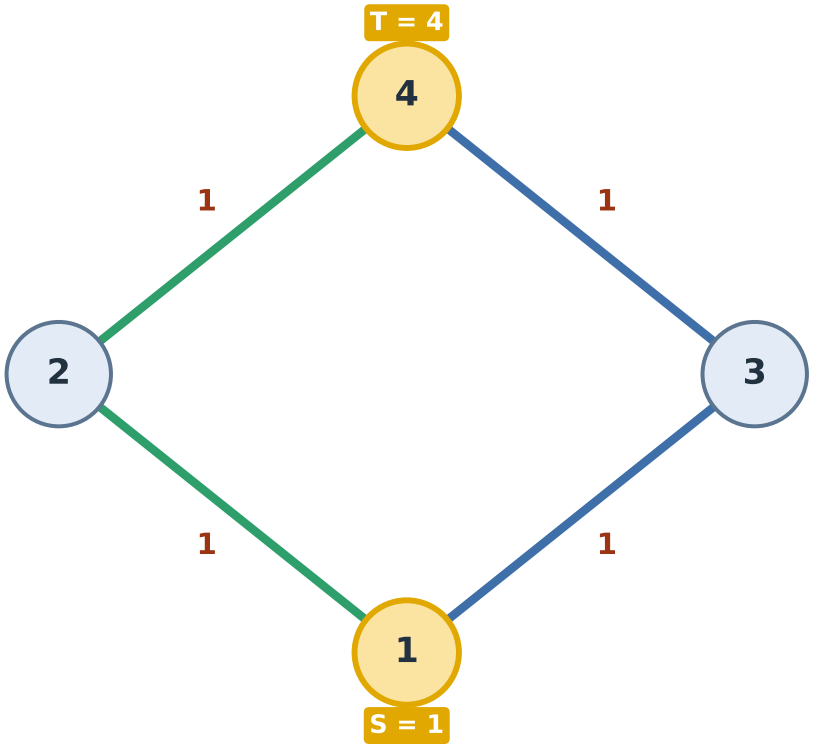
The red loop is a negative self-loop (weight −1) on vertex 4. Vertex 4 cannot be reached from S = 1, so the loop does not matter.

Walkthrough

From 1: dist(2) = 4 and dist(3) = 4 + (−2) = 2. No edge leads into vertex 4, so it stays INF; the final check finds no improving edge from a reachable vertex, so there is no reachable negative cycle.

M19-17 — Count Shortest Paths Modulo P

Difficulty: Hard | Topic: Graphs



Sample input

```
4 4 1 4 1000000007
1 2 1
2 4 1
1 3 1
3 4 1
```

Sample output

```
2
```

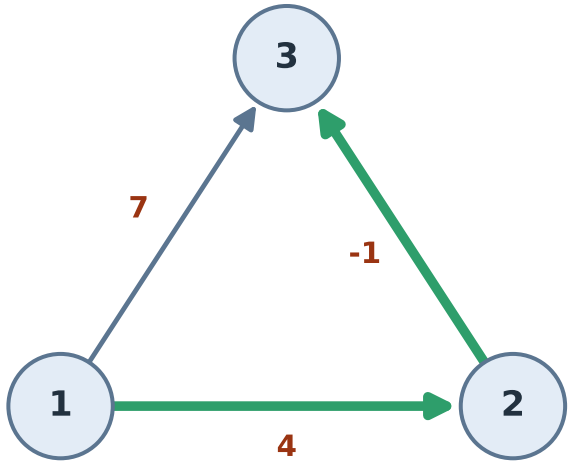
Numbers are edge weights. The two colors show the two different shortest routes from S = 1 to T = 4.

Walkthrough

Both routes have total weight 2. Dijkstra reaches 4 once through 2 and again with the same distance through 3, so it adds the path counts: 1 + 1 = 2.

M19-18 — All-Pairs Shortest Paths (Floyd–Warshall)

Difficulty: Hard | Topic: Graphs



distance matrix

```
from 1:  0 4 3

from 2:  INF 0 -1

from 3:  INF INF 0
```

Sample input

```
3 3
1 2 4
2 3 -1
1 3 7
```

Sample output

```
0 4 3
INF 0 -1
INF INF 0
```

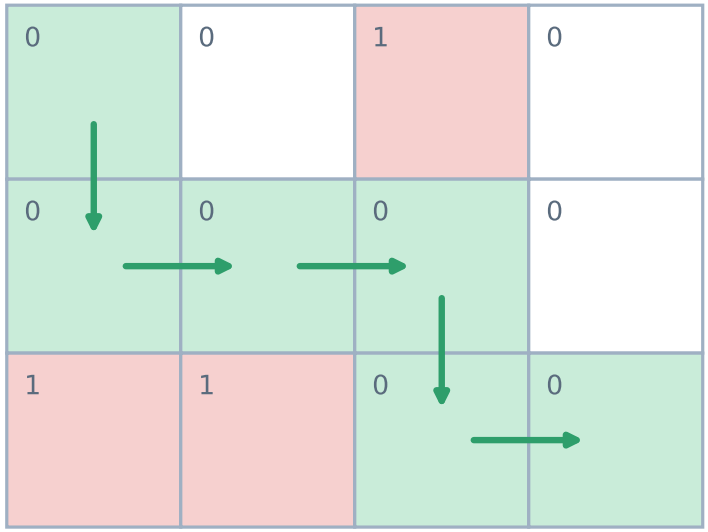
Green arrows: the shortest route from 1 to 3 (through vertex 2). The matrix lists every ordered pair; INF means no route.

Walkthrough

1 → 2 → 3 costs 4 + (−1) = 3, which beats the direct edge of 7. No edge enters vertex 1 and no edge leaves vertex 3, so every entry below the diagonal is INF.

M19-19 — Shortest Path Through a Grid with Obstacles

Difficulty: Medium | Topic: Graphs



Sample input

```
3 4
0 0 1 0
0 0 0 0
1 1 0 0
```

Sample output

```
5
```

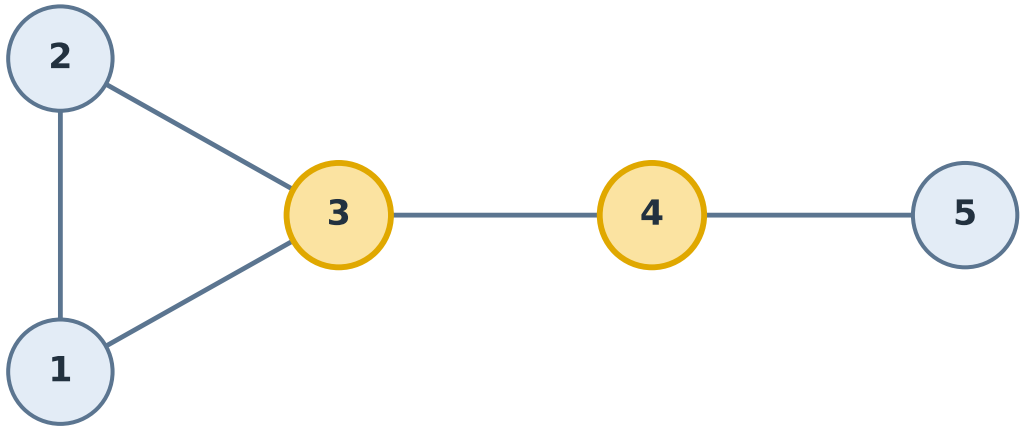
Red cells are blocked. Green cells and arrows show one shortest route from the top-left to the bottom-right cell.

Walkthrough

Down, right, right, down, right reaches the goal in 5 moves. BFS from the top-left cell assigns each open cell its distance; the bottom-right cell gets 5.

M19-20 — Find Articulation Points

Difficulty: Hard | Topic: Graphs



articulation points

Gold vertices are articulation points: removing either one (with its edges) splits the graph.

Sample input

```
5 5
1 2
2 3
3 1
3 4
4 5
```

Sample output

```
2
3 4
```

Walkthrough

Removing 3 leaves {1, 2} and {4, 5} disconnected; removing 4 isolates 5. Removing 1, 2, or 5 leaves everything else connected.